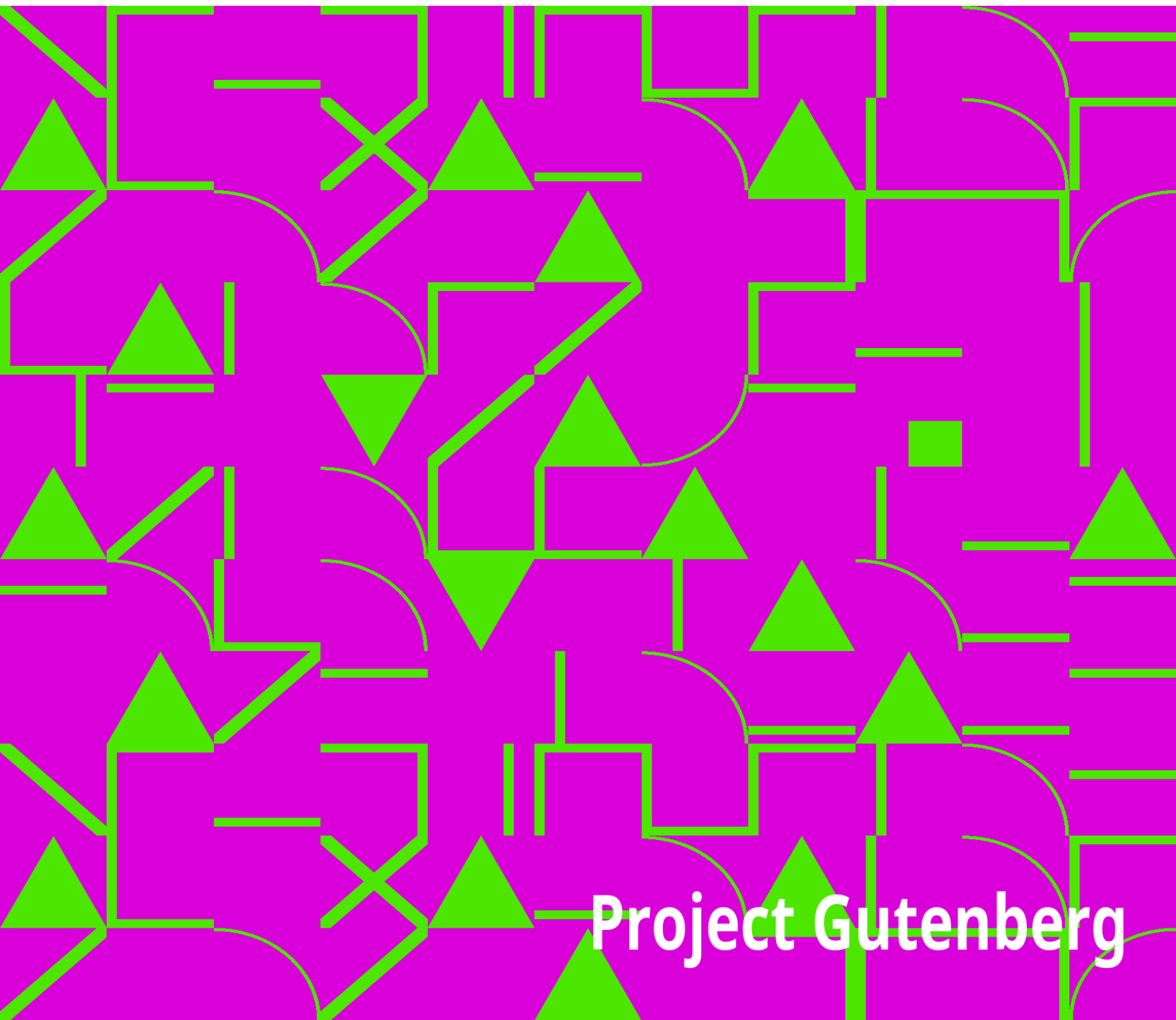


Morphological Variation in a Population of the Snake, *Tantilla gracilis* Baird and Girard

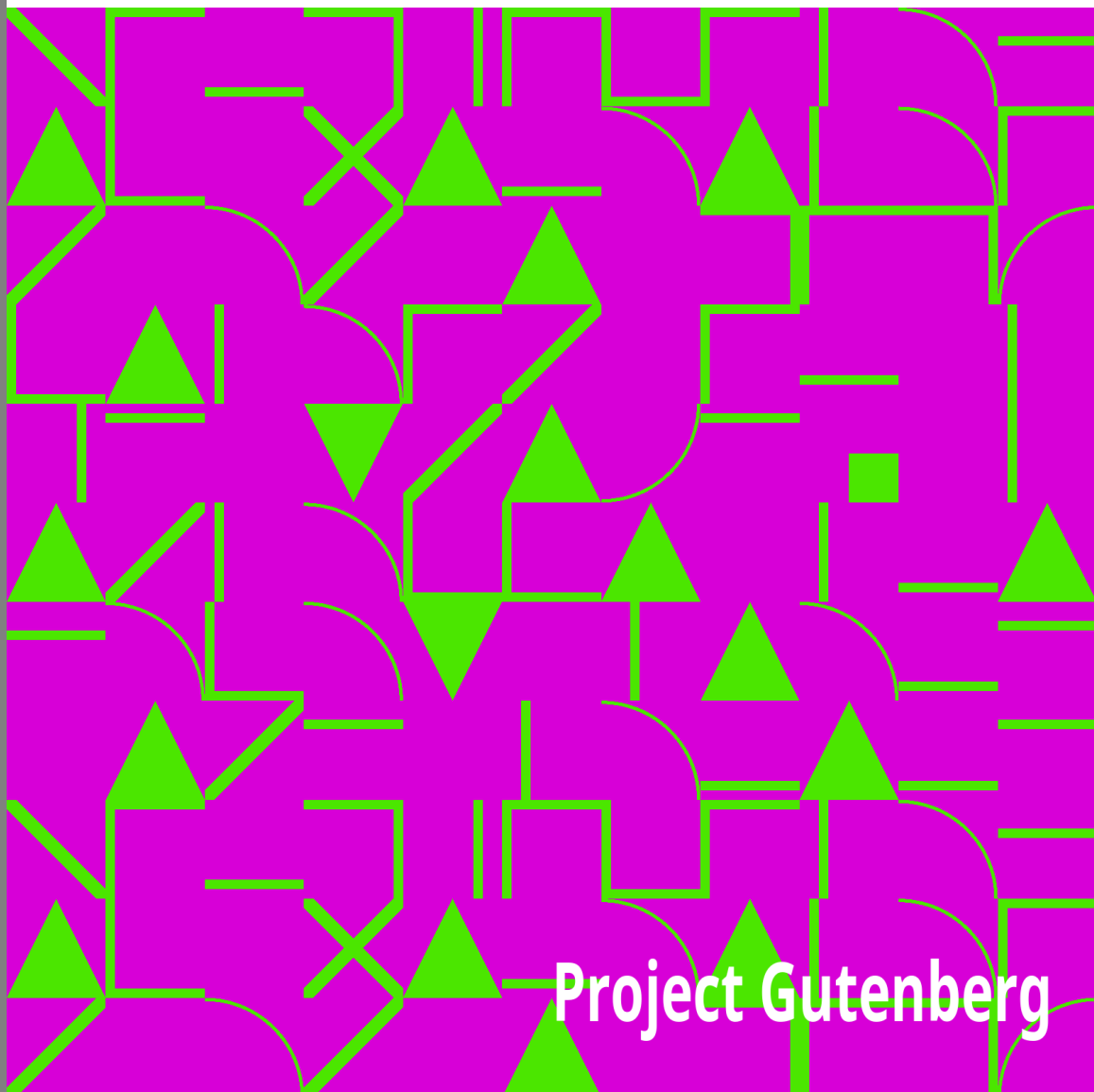
Laurence M. Hardy and Charles J. Cole

An abstract geometric pattern consisting of various shapes including squares, triangles, circles, and lines, all in shades of green and yellow, creating a complex, tessellated background.

Project Gutenberg

Morphological Variation in a Population of the Snake, *Tantilla gracilis* Baird and Girard

Laurence M. Hardy and Charles J. Cole



Project Gutenberg

The Project Gutenberg eBook of Morphological Variation in a Population of the Snake, *Tantilla gracilis* Baird and Girard

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Morphological Variation in a Population of the Snake, *Tantilla gracilis* Baird and Girard

Author: Laurence M. Hardy
Charles J. Cole

Release date: October 24, 2010 [eBook #34127]
Most recently updated: January 7, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/34127

Credits: Produced by Simon Gardner, Chris Curnow, Joseph Cooper and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK MORPHOLOGICAL
VARIATION IN A POPULATION OF THE SNAKE, TANTILLA GRACILIS BAIRD
AND GIRARD ***

Transcriber's Notes

The scale of [Figure 4](#) and [Figure 5](#) are preserved from the original, if viewed on a 96 DPI monitor screen and with a browser setting of 100%.

The only changes made to the text are the following:

Page 623 Table 3 column heading: changed "or" to "of" (Number of specimens).

Page 625: changed "percent" to "per cent" (92 per cent of the time).

Table of Contents:

[Introduction](#)

[Results](#)

[Literature Cited](#)

UNIVERSITY OF KANSAS PUBLICATIONS
MUSEUM OF NATURAL HISTORY

Volume 17, No. 15, pp. 613-629, 6 figs.
May 14, 1968

Morphological Variation in a Population of the Snake, *Tantilla gracilis* Baird and Girard

BY

LAURENCE M. HARDY AND CHARLES J. COLE

UNIVERSITY OF KANSAS
LAWRENCE
1968

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, Frank B. Cross, Henry S. Fitch, J. Knox Jones,
Jr.

Volume 17, No. 15, pp. 613-629, 6 figs.
Published May 14, 1968

University of Kansas
Lawrence, Kansas

PRINTED BY
ROBERT R. (BOB) SANDERS, STATE PRINTER
TOPEKA, KANSAS
1968

31-9422

Morphological Variation in a Population of the Snake,
Tantilla gracilis Baird and Girard

By

LAURENCE M. HARDY AND CHARLES J. COLE

INTRODUCTION

Variation of selected taxonomic characteristics of flat-headed snakes (*Tantilla gracilis* Baird and Girard) from several midwestern states was studied by Force (1935), but she did not attempt a comprehensive evaluation. Taylor (1936) reported on variation in *T. gracilis* from various localities in Kansas; Kirn, Burger, and Smith (1949) studied selected structures of specimens of *T. gracilis* from throughout its range. The present study was undertaken to determine the variation in both currently used and potential taxonomic characteristics of *T. gracilis* from one locality.

We are thankful to Charles W. Myers, Gorgas Memorial Laboratory, for suggestions concerning characteristics examined. We are indebted to Drs. William G. Degenhardt, University of New Mexico, Herndon G. Dowling, New York Zoological Society, Charles H. Lowe, University of Arizona, and Richard G. Zweifel, American Museum of Natural History, for criticizing the manuscript. Dr. William E. Duellman permitted us to study specimens in the University of Kansas Museum of Natural History.

Materials and Methods

The specimens examined were donated to the Museum of Natural History, University of Kansas, by the late Paul Anderson of Independence, Missouri. All specimens (KU numbers 83435-83680; N = 246) were collected in the vicinity of Winfield, Cowley County, Kansas, by Charles E. Burt and students from Southwestern College in the period from 1938 to 1941, inclusive.

Both authors gathered data on most characteristics and examined each atypical individual. Hardy determined characters of the maxillae (which were removed; N = 20), body and total lengths, and the scale formula of KU 83620.

We examined 22 characteristics of external morphology and 10 of the maxillae. All paired characteristics (excepting those of the maxillae) were examined on both sides of each snake and are referred to from the left side to the right side (for example, "preoculars 1-2" means there is one preocular on the left side and there are two on the right). Oviducts and oviducal eggs were observed on specimens designated as females; hemipenes were examined on specimens designated as males. Total length was measured by straightening each snake along a 300 mm. scale. Tail length was measured only on snakes having complete tails, and body length (snout-vent) was

considered as total length minus tail length. The body length was measured on snakes having incomplete tails.

Supralabials, infralabials, preoculars and postoculars were counted, and we followed Peters (1960:9) in determining the nature of the temporals. Occasionally the primary (anterior) temporal is separated from the postocular by contact of the parietal and a supralabial (Taylor, 1936:338). We followed Peters (1964:219-220) in determining characteristics of the nasal and we looked for contact of the prefrontal with particular supralabials (Taylor, 1936:338). We determined which supralabials are in contact with the eye and whether the mental is in contact with the anterior pair of chin-shields.

The number of rows of dorsal scales was determined at five locations: a) around the neck—counts were made from each side diagonally and posteriorly from the anteriormost dorsal scale in contact with the first ventral (see below)—counts from the left and right sides are the first and second counts, respectively, recorded in the scale formula; b) one head length posterior to head—counts were made either diagonally and posteriorly or in a zig-zag pattern, beginning at the end of the second count around the neck—this is the third count recorded in the scale formula; c) midbody (approximated)—counts were made either diagonally and posteriorly or in a zig-zag pattern—this is the fourth count in the scale formula; d) anterior to anus (preanal)—counts were made in a zig-zag pattern beginning on one side and ending on the other at the first dorsal scale in contact with the anal plate—this is the fifth count in the scale formula.

We followed the method of Dowling (1951:98-99) in counting ventrals, and the anal plate was determined as entire, divided, or partly divided. Subcaudals were counted on only the right side beginning at the first scale that contacts a corresponding scale from the opposite side; fusion of particular pairs of subcaudals was determined. The "spine" at the tip of the tail was not counted. The head scutellation of each specimen was examined for abnormalities.

We determined the number of fangs and the number of maxillary teeth anterior to them and recorded the presence or absence of a diastema between the anterior fang and the maxillary tooth immediately preceding it. The diastema, if present, is a space distinctly wider than the spaces separating the prediastemal teeth.

The point of intersection of two perpendicular lines of an ocular grid determined which tooth-socket occurs directly opposite the distal tip of the suborbital process ([Fig. 5b](#)), but if that point occurs between two sockets then the number recorded was the number of the tooth immediately anterior to that point plus one-half.

The anterior and posterior edges of the suborbital process were determined as parallel or not at any point by aligning them with parallel lines on an ocular grid ([Fig. 5b](#)).

The angle of the posterior edge of the lateral flange was determined as an acute, obtuse, or right angle by fitting it to a pair of perpendicular lines of an ocular grid ([Fig. 5b](#)).

The number of lateral anterior foramina ([Fig. 4b](#)) was determined. If only two are present, their lengths (anterior to posterior) were compared by measuring them with an ocular micrometer. Also, if two lateral anterior foramina are present, the point of intersection of two perpendicular lines of an ocular grid determined which tooth-socket occurs directly below the center of the posterior foramen ([Fig. 4b](#)).



RESULTS

General Characteristics

From the data given by Force (1935), Taylor (1936), and Kirn *et al.* (1949), and from our own data, we conclude that a typical specimen of *gracilis*, meaning a specimen having the usual characteristics of the species, has the following characteristics in combination: supralabials 6-6; infralabials 6-6; preoculars 1-1; postoculars 1-1; temporals 1 + 1 on each side; nasal divided below naris; supralabials 3 + 4 entering orbit; mental in contact with chin-shields; all five dorsal scale counts 15; and plate divided. (In the present work we use the words usual, unusual, typical, and atypical with neither quotation marks nor apology).

Of the 244 specimens on which we could examine each of these characteristics, only 125 (51.2%) have all of the typical characteristics in combination. The 119 specimens that do not possess all of the typical characteristics in combination include 31 (12.7% of the 244) that lack at least two of the typical characteristics.

Sex

The sample (N = 246) is composed of 107 (43.5%) females and 139 (56.5%) males.

Measurements

Females have total lengths from 96 to 244 mm. (mean, 173.3; N = 79). One female (KU 83480) measuring 244 mm. may be the longest specimen known (Conant, 1958, reports the longest as 9-1/8 inches—approximately 232 mm.). Males have total lengths from 96 to 215 mm. (mean, 162.4; N = 109). Females have tail lengths from 16 to 50 mm. (mean, 34.6; N = 79), and males have tail lengths from 21 to 53 mm. (mean, 37.8; N = 109).

The ratio of tail length to total length in females is from 0.17 to 0.22 (mean, 0.20; N = 79); in males it is from 0.21 to 0.27 (mean, 0.23; N = 109; [Fig. 1](#)). Juveniles (those less than 125 mm. in total length according to Force, 1935: tables 1 and 2) have proportionally shorter tails than do adults; the ratio of tail length divided by total length is 0.17-0.21 (mean, 0.18; N = 14) in females and 0.21-0.23 (mean, 0.22; N = 16) in males. Adult females (125 mm. or more in total length) have ratios from 0.18 to 0.22 (mean, 0.20; N = 65) and adult males have ratios from 0.21 to 0.27 (mean, 0.23; N = 93).

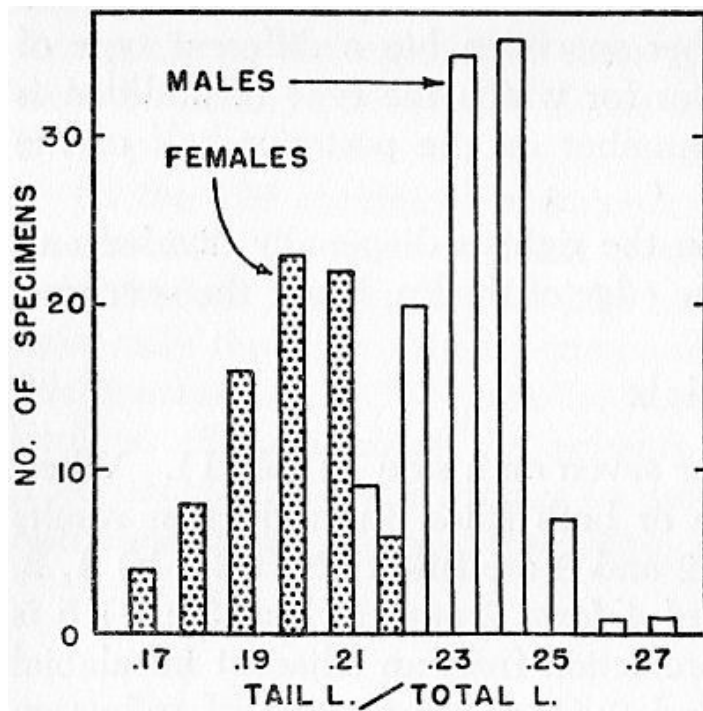


Fig. 1. Ratio of tail length to total length in females and males of *T. gracilis*.

Of the 246 specimens examined, 58 have incomplete tails. Of these 58 specimens, six had freshly broken tails, so we assume that 52 (21.1% of the 246) have incomplete tails resulting from natural causes; the remaining six specimens could have had their tails broken at or soon after capture, assuming that none of the snakes was retained in captivity for an extended period.

Supralabials

The supralabials are either six (usually) or seven on a side ([Table 1](#)). When seven supralabials are present on either one or both sides, the addition results from one of the following conditions: a) 5 is divided to form 5 and 6 (N = 6); b) 2 and 3 form 2, 3, and 4 (N = 1); c) 5 and 6 form 5, 6, and 7 (N = 12); d) 4, 5, and 6 form 4, 5, 6, and 7 (N = 1); e) 1 and 2 form 1, 2, and 3 (N = 1); f) 1, 2, and 3 form 1, 2, 3, and 4 (N = 1); g) 2 is fused with the nasal, and 3 and 4 are reduced to granules (N = 1); h) type of addition unknown (N = 5).

TABLE 1. Variation in Number of Supralabials (N = 245) and Infralabials (N = 246) of *Tantilla gracilis*.

Number of Supralabials	Number of specimens	Per cent of occurrence	Number of infralabials	Number of specimens	Per cent of occurrence
6-6	221	90.2	5-5	5	2.0
6-7	10	4.1	5-6	4	1.6
7-6	10	4.1	6-5	5	2.0
7-7	4	1.6	6-6	229	93.1
			6-7	1	0.4

In three of the four specimens with 7-7 supralabials, the type of addition is the same on each side, whereas the other specimen has a different type of addition on each side. Of the 23 examples for which the type of addition is known, 19 (82.6%) have the increase in number on the posterior half of the series.

In one specimen the third supralabial on the right is diagonally divided and the resulting upper part does not contact the edge of the lip; hence the specimen has 6-6 supralabials.

Infralabials

The infralabials are five, six (usually), or seven on a side ([Table 1](#)). When five infralabials are present on either one or both sides, the reduction results from one of the following conditions: a) 2 and 3 are fused (N = 3); b) 1, 2, and 3 form 1 and 2 (N = 6); c) 2, 3, and 4 form 2 and 3 (N = 3); d) 5 is separated from the edge of the lip by a projection from an adjacent infralabial (N = 1); e) 1, 2, 3, and 4 form 1, 2, and 3 (N = 2); f) type of reduction unknown (N = 4).

Of the specimens with 5-5 infralabials, two have the same type of reduction on each side, one has a different type of reduction on each side, and the type of reduction is unknown for the other two specimens.

When seven infralabials are present on either one or both sides, the addition can result from infralabials 2 and 3 forming 2, 3, and 4 (N = 2); the type of addition is unknown in one specimen.

In one specimen the second infralabial on the right side is reduced to a small wedge-shaped granule and in another specimen several infralabials on both sides are irregularly divided horizontally, although both specimens have six infralabials on the sides having the abnormalities.

Preoculars

Of the 245 specimens examined, 242 (98.8%) have one preocular on each side. Both preoculars are absent from two specimens, and one specimen has 1-0. In each case when a preocular is lacking, the "preocular" and prefrontal are fused. One specimen having 1-1 preoculars has both of them reduced to small vertical slivers.

Postoculars

Of the 245 specimens examined, 242 (98.8%) have one postocular on each side. Postoculars are 2-1 in two specimens and 1-2 in one. In one specimen the lower of the two postoculars on each side is very small, being approximately 1/6 the diameter of the upper one (measured with an ocular micrometer).

Temporals

Of the 245 specimens examined, 235 (95.9%) have 1 + 1 temporals on each side. Four specimens exhibit addition of temporals: one specimen has two secondary temporals on the left, and one has a tertiary temporal on the left. Two specimens have partly healed head injuries that probably caused atypical temporal conditions: one specimen has two primary temporals on the left, and one has two secondary temporals on the left and two upper and one lower secondary temporal on the right.

Six specimens exhibit reduction of temporals: one lacks a primary temporal on the left ("temporal" is fused with sixth supralabial), two lack secondary temporals on the right, and three lack secondary temporals on the left. The right secondary "temporal" of one specimen is separated from the supralabials by a posterior extension of the primary temporal.

Of the 246 specimens examined, 200 (81.3%) have both primary temporals in contact with their adjacent postoculars. Thirteen specimens (5.3%) have only the left primary temporal in contact with the postocular and eight (3.3%) have only the right primary temporal in contact with the postocular. Twenty-five specimens (10.2%) have both primary temporals separated from the postoculars. Separation of the primary temporal from the postocular, whenever it occurs, results from contact of mutual extensions from the adjacent parietal and supralabial (usually the fifth).

Nasal

Of the 243 specimens examined, 237 (97.5%) have the nasal divided below the naris on both sides. Three specimens have the nasal completely divided (above and below the naris) on only the left side; one specimen has a completely divided nasal on only the right side. A groove, but not a complete division of the nasal, is present above the left naris of one specimen. One specimen lacks the typical division below the naris on both sides, but a complete suture is present posterior to the naris and the resulting postnasal scales are, according to position, loreals.

Contact of prefrontal with supralabials

Of the 243 specimens examined, 147 (60.5%) have the prefrontal separated from the labials on each side. Of the remaining 96 specimens, 53 (21.8% of the 243) have the

prefrontals in contact with the second labial on each side; five other variations of this characteristic were observed ([Table 2](#)).

TABLE 2. Variation in Contact of the Prefrontals and the Labials in 243 Specimens of *Tantilla gracilis*.

CHARACTERISTIC	Number of specimens	Per cent of occurrence
No contact	147	60.5
Contact of prefrontal with second labial on right	26	10.7
Contact of prefrontal with second labial on left	10	4.1
Contact of prefrontal with each second labial	53	21.8
Contact of prefrontals with labial two on left and labial three on right	1	0.4
Contact of prefrontals with labial three on left and labial two on right	3	1.2
Contact of prefrontals with labial two on left and labials two and three on right	3	1.2

The specimen having the prefrontal in contact with labial 2 on the left and labial 3 on the right has seven supralabials on the right, the extra labial being added anterior to the eye. In two of the three specimens with the prefrontals in contact with labial 3 on the left and labial 2 on the right, the preoculars and prefrontals are fused on each side; the third specimen has each preocular reduced to a small vertical sliver. Of the three specimens having the prefrontals in contact with labial 2 on the left and labials 2 and 3 on the right, two are typical with respect to other characteristics that might result in this atypical condition, but one has the prefrontals and preoculars fused.

Supralabials entering orbit

Of the 246 specimens examined, 233 (94.7%) have the third and fourth supralabials entering each eye ([Table 3](#)). All snakes that have the fourth and fifth labials entering the orbit on either side also have seven supralabials on the same side. One specimen having only the fourth labial entering the eye on the right side also has six supralabials on that side, but the third supralabial is split diagonally and the part contacting the eye does not contact the lip and hence is not a supralabial. The remaining three atypical conditions (4/3 + 4; 3 + 4/3 + 4 + 5; 3 + 4 + 5/3 + 4) occur only one time each and on snakes having 6-6 supralabials.

Contact of mental with chin-shields

Of the 246 specimens examined, 190 (77.2%) have the mental in contact with both anterior chin-shields. The mental is in contact with only the left anterior chin-shield in 15 specimens, only the right anterior chin-shield in 2 specimens, and the mental does not contact either anterior chin-shield in 39 specimens. Separation of the mental and an anterior chin-shield always results from the posterior elongation of a first infralabial.

TABLE 3. Variation in Number of Supralabials Entering the Eye in 246 Specimens of *Tantilla gracilis*.

SUPRALABIALS		Number of Per cent of	
Left	Right	specimens	occurrence
3 + 4	3 + 4	233	94.7
4 + 5	3 + 4	4	1.6
4	3 + 4	1	0.4
3 + 4	4 + 5	5	2.0
3 + 4	4	1	0.4
3 + 4	3 + 4 + 5	1	0.4
3 + 4 + 5	3 + 4	1	0.4

Dorsal scale rows

Of the 245 specimens examined for this characteristic, 228 (92.7%) have scale counts of 15-15-15-15-15. The number of scale rows around the neck and one head length behind the head are 15 except in a single specimen that has 14 scale rows at these three places. The number of scale rows around midbody is 15 in 244 of 246 specimens (99.2%); two specimens have 14 scale rows around midbody. The number of preanal scale rows is more variable; counts of 14 through 17 were recorded. Fourteen specimens have 14 preanal scale rows (5.7%), 229 have 15 preanal rows (93.1%), two have 16 preanal rows (0.8%), and one has 17 preanal rows (0.4%). Known causes for atypical numbers of preanal scale rows are listed in [Table 4](#). A preanal count at one head length anterior to the anus would have omitted most of the above variation in numbers of preanal scale rows.

TABLE 4. Atypical Numbers of Preanal Scale Rows in Specimens of *Tantilla gracilis*.

Preanal scales	Number of specimens	CAUSE	Per cent of occurrence
14	10	Scale rows seven and eight on left fuse within the last five ventrals	4.1

14	1	Scale row two on the right is lost adjacent to the penultimate ventral	0.4
16	1	Scale row eight on the left divides adjacent to the last ventral	0.4
16	1	A scale row is added on the right adjacent to and anterior to the anal plate	0.4
17	1	A scale row is added on each side adjacent to and anterior to the anal plate	0.4

The complete dorsal scale formula for KU 83620, a male, is as follows:

$$\begin{array}{ccccccc}
 15 \frac{6+7(2)}{6+7(2)} & 13 \frac{+7(4)}{+7(5)} & 14 \frac{+7(11)}{6+7(11)} & 13 \frac{+7(17)}{6+7(17)}, \\
 & 14 \frac{+7(18)}{6+7(18)} & 13 \frac{+7(25)}{+7(25)}, \\
 14 \frac{+7(26)}{+7(26)} & 15 \frac{+7(36)}{6+7(36)} & 14 \frac{+7(38)}{6+7(38)} & 13 \frac{+7(39)}{+7(42)} & 14 \frac{+7(42)}{+7(42)}, \\
 & 15 \frac{+7(45)}{6+7(45)} & 14 \frac{+7(53)}{+7(53)}, \\
 15 \frac{7+8(67)}{7+8(67)} & 14 \frac{+7(69)}{+7(69)} & 15 \frac{+7(90)}{6+7(90)} & 14 \frac{+7(93)}{+7(93)} & 15 \frac{+7(99)}{6+7(99)}, \\
 & 14 \frac{+7(100)}{+7(100)} & 15(122).
 \end{array}$$

[\[Textual representation of dorsal scale formula.\]](#)

Specimen 83620 is the only one for which we have determined the complete dorsal scale formula. Obviously the formula is complex, and possibly the specimen is atypical in this regard. Nevertheless, addition and reduction of scale rows is apparent on other specimens and can be easily observed by examining specimens under a dissection microscope.

Anal plate

Of the 246 specimens examined, 232 (94.3%) have the anal plate divided. Of the 14 specimens having undivided anal plates, four have an incomplete groove anteriorly in the plate.

Ventrals

Females have from 122 to 137 (mean, 130.8 ± 0.26 ; $N = 107$) ventrals; males have from 117 to 131 (mean, 122.3 ± 0.21 ; $N = 139$; [Fig. 2](#)). Of the 246 specimens examined, 29 (11.8%; 11 females and 18 males) have an added half-ventral immediately anterior to the anal plate. Of these specimens, 27 have the added half-ventral on the left side and two have it on the right. Five specimens have the last ventral divided similar to the normal division of the anal plate. One specimen has the last ventral incompletely divided transversely. Excluding the last ventral, seven specimens have divided ventrals (half-ventrals). Four of these specimens have added half-ventrals on the right side (two specimens with one half-ventral each, one specimen with two half-ventrals, and one specimen with three half-ventrals), and four have added half-ventrals on the left side (one half-ventral on each specimen).

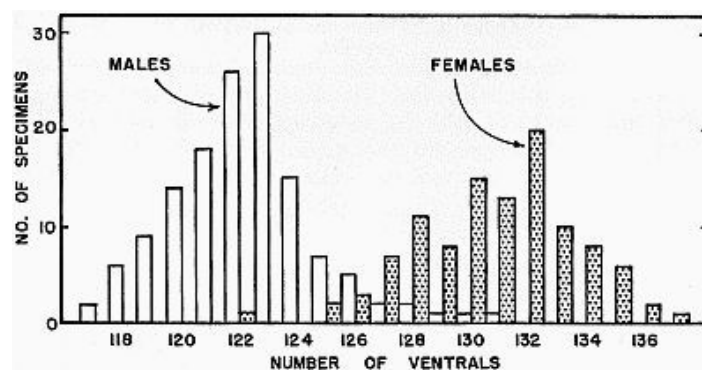


Fig. 2. Number of ventrals in females and males of *T. gracilis*.

Subcaudals

Females have from 37 to 47 (mean, 42.0 ± 0.20 ; $N = 79$) subcaudals; males have from 43 to 55 (mean, 48.4 ± 0.23 ; $N = 109$; [Fig. 3](#)). Of the 187 specimens examined, 13 (7.0%) have the last pair of subcaudals fused. One specimen has left subcaudal 28 fused with both adjacent subcaudals (also fused) on the right. While reading proof we found two omissions in [Fig. 3](#); there should be illustrated two males with 43 subcaudals and three males with 44 subcaudals.

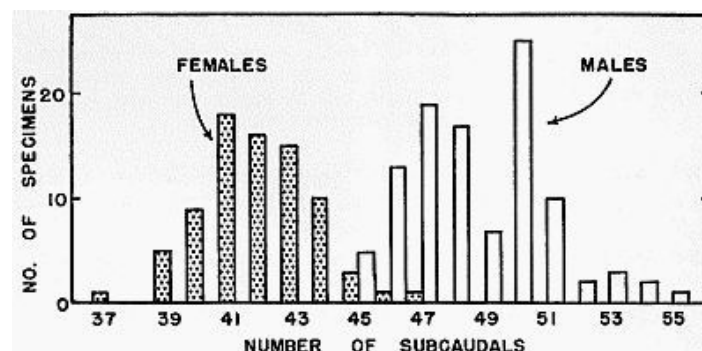


Fig. 3. Number of subcaudals in females and males of *T. gracilis*.

Ventrals minus subcaudals

The value for number of ventrals minus the number of subcaudals varies from 80 to 96 (mean, 88.8 ± 0.39) for 79 females, and from 67 to 83 (mean, 73.8 ± 0.28) for 109 males.

Additional scale characteristics

Of the 246 specimens examined, four (1.6%) have the rostral in contact with the left prefrontal, and hence the internasals are separated from one another. One snake has approximately one-fourth (anteriorly) of the suture between the parietals fused. One specimen has the posterior edge of the frontal fused with the left parietal. One specimen has the parietal shield on the right side abbreviated posteriorly. One specimen has both prefrontals fused with the frontal, although an incomplete suture is present.

Maxillary characteristics (Figs. [4](#) and [5](#))

Maxillary teeth anterior to fangs 10-14 (mean, 12.1; N = 20); tooth-socket number 6.5-8.5 (mean, 7.2; N = 20) directly opposite distal tip of suborbital process; lateral anterior foramina 1-3 (mean, 1.8; N = 20); tooth number 6-9 (mean, 6.9; N = 12) directly below center of posterior foramen (if only two foramina present). Three specimens lack a diastema and 17 specimens have a diastema. The anterior and posterior edges of the suborbital process are parallel in two specimens and not parallel in 14; the process is broken in four specimens. The lengths (anterior to posterior) of the lateral anterior foramina (if only two are present) are approximately equal in six specimens, and in six the anteriormost one is the longest; six specimens have one foramen and two specimens have three foramina. All specimens examined have two fangs, with the posterior edge of the lateral flange forming an obtuse angle.

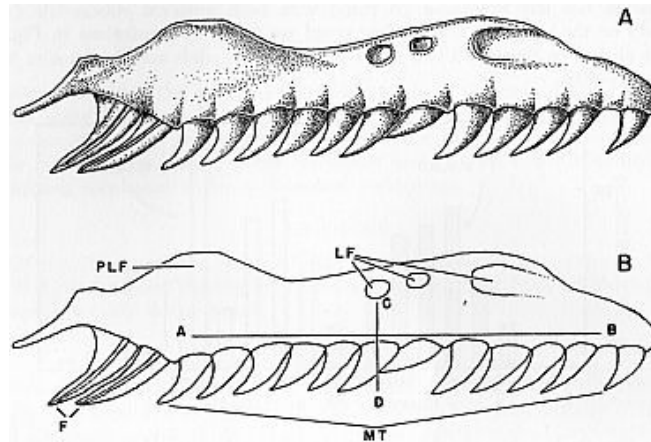


Fig. 4. Lateral view of right maxilla of *T. gracilis* (KU 83484), $\times 39$. Actual length of bone is 2.44 mm. A. Detail of bone. B.

Outline sketch of bone to illustrate characteristics examined. Lateral anterior foramina (LF); fangs (F); "prediastemal" maxillary teeth (MT); lateral flange (PLF); perpendicular lines, A-B and C-D, are used to determine which tooth-socket occurs directly below the center of the posterior foramen (the specimen illustrated is atypical because there are three foramina present).

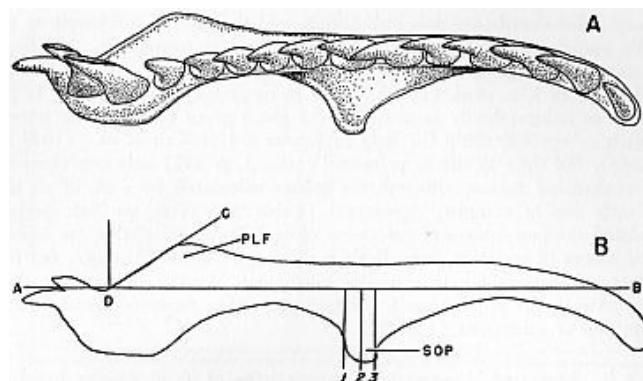


Fig. 5. Ventral view of right maxilla of *T. gracilis* (KU 83484), $\times 39$. Actual length of bone is 2.44 mm. A. Detail of bone. B.

Outline sketch of bone to illustrate characteristics examined. Lateral flange

(PLF) forms an obtuse angle (ADC) with main axis of bone (AB); a reference line (2), perpendicular to the main axis of the bone, is used to locate the tooth-socket directly opposite the tip of the suborbital process (SOP); the anterior and posterior edges of the suborbital process are not parallel at any point (reference lines 1 and 3).

Discussion and Conclusions

Generic characteristics of *Tantilla* are as follows: size small, body slender, head not noticeably distinct from body; preoculars one; postoculars one or two; nasals two; loreal absent; smooth dorsal scales in 15 rows; anal plate usually divided but sometimes single; subcaudals paired; usually two enlarged, posterior, grooved fangs on the maxilla separated by a small diastema from the other maxillary teeth (Baird and Girard, 1853:131; Cope, 1900:1110; Dunn, 1928:24; Blanchard, 1938:369; Schmidt and Davis, 1941:268; Wright and Wright, 1957:722; Conant, 1958:180). Of these characteristics, only the following were invariable in our sample: size small, body slender, head not noticeably distinct from body; postoculars one or two; smooth dorsal scales; posterior grooved fangs on maxilla. The dorsal scales, anal, and subcaudals were "correct" 92 per cent of the time; the preoculars and nasals were "correct" 97 per cent of the time.

Tantilla reportedly has two nasals (Baird and Girard, 1853:131; Cope, 1900:1110; Wright and Wright, 1957:722). *Tantilla gracilis* usually has a single nasal that is divided below the naris; variation in this characteristic is discussed above.

In addition to the generic characteristics, specific characteristics for *T. gracilis* are as follows: supralabials 5-8 (usually 6); supralabials 3 and 4 entering orbit; infralabials 5-7 (usually 6); temporals 1 + 1; ventrals 115-138 in females, 106-132 in males; subcaudals 33-53 in females, 40-57 in males; tail length 13-27 per cent total length in females, 15-30 per cent in males (Baird and Girard, 1853:132; Cope, 1900:1111-12; Force, 1935:653-54; Taylor, 1936:337-38; Blanchard, 1938:371-72; Kirn, Burger, and Smith, 1949:240-49). Excepting the number of temporals and the supralabials entering the orbit, the characteristics of the specimens in our sample are within the ranges of variation mentioned above. Of the 10 maxillary characteristics studied, no variation was observed in number of fangs or angle of lateral flange.

Because our data have some bearing on the problem of geographic variation in *T. gracilis* and the recognition of subspecies, we comment briefly on the status of subspecies in *T. gracilis*. Kirn, Burger and Smith (1949) proposed the recognition of two subspecies of *Tantilla gracilis* (*T. g. gracilis* Baird and Girard and *T. g. hallowelli* Cope). These subspecies were diagnosed on the basis of differences in ventrals, subcaudals, and the ratio of tail length to total length; sexual dimorphism in each characteristic was considered. We do not recognize these subspecies for the reasons given below.

The data of Kirn *et al.* (1949) appear to be presented inaccurately in part. Both of us independently recalculated the mean given for each characteristic for each subspecies, using the data in figures 2-4 of Kirn *et al.* (1949:242, 244-245). Of their 12 means presented (table 1, p. 247) only two agree with our recalculated means, although the means calculated by each of us independently are in complete agreement ([Table 5](#)). Also, we independently calculated the percentages of specimens of each "subspecies" that are included in the ranges of variation given in their diagnoses ([Table 5](#)); again, our independent calculations are in complete agreement. In our opinion the differences between the populations for the characteristics analyzed do not warrant recognition of subspecies ([Fig. 6](#)).

TABLE 5. Means of "Diagnostic" Characteristics of *T. g. gracilis* Baird and Girard and *T. g. hallowelli* Cope.

CHARACTERISTICS	<i>T. g. gracilis</i>			<i>T. g. hallowelli</i>		
	Mean (Kirn, <i>et al.</i> 1949)	Mean (Kirn, <i>et al.</i>) recalculated by us	Per cent specimens of Kirn, <i>et al.</i> included their diagnosis	Mean (Kirn, <i>et al.</i> in 1949)	Mean (Kirn, <i>et al.</i>) recalculated by us	Per cent specimens of Kirn, <i>et al.</i> included in their diagnosis
Ventrals— females	125.67	125.71	69.4	130.07	130.07	79.6
Ventrals—males	115.97	116.61	70.0	121.22	120.87	69.7
Caudals— females	40.99	40.82	62.8	46.79	43.82	77.4
Caudals—males	47.75	48.29	78.3	51.67	50.29	51.0
Tail L./total L. females	.1976	.1976	69.0	.2084	.2076	74.2
Tail L./total L.						

males .2336 .2362 56.1 .2477 .2423 79.6

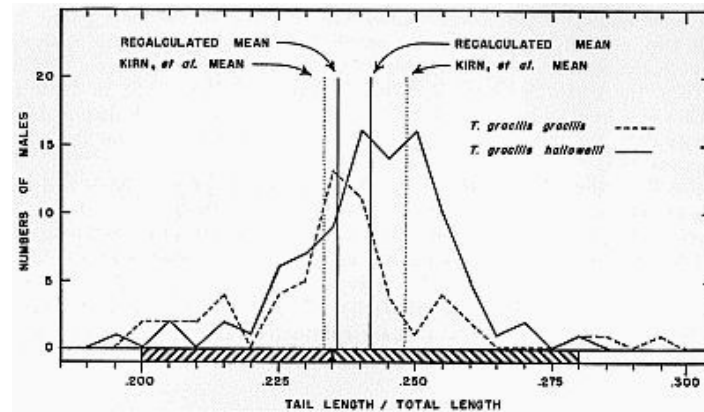


Fig. 6. Frequency polygons presenting ratios of tail length to total length for males of *T. g. gracilis* and *T. g. hallowelli* from Kirn, et al. (1949). The dotted vertical lines represent the means given by Kirn, et al. (1949) for *gracilis* (left) and *hallowelli* (right); the solid vertical lines represent the recalculated means for *gracilis* and *hallowelli*, respectively, using the data of Kirn, et al. The hatched bars represent the range included by Kirn, et al. (1949) in their diagnoses of *gracilis* (left) and *hallowelli* (right).

The data presented by Kirn *et al.* do not demonstrate intergradation between two populations. Moreover, the diagnostic ranges of the number of subcaudals in males of *hallowelli* and the ratio of tail length to total length in males of *gracilis* do not include the recalculated means for those characteristics. Furthermore, the means for the following characteristics are dangerously close to being excluded from their respective "diagnostic" ranges: in *hallowelli*, the number of ventrals in males; in *gracilis*, the number of subcaudals in females, and the ratio of tail length to total length in females. It is incongruous that Kirn *et al.* state (p. 243) that "the coefficient of geographic divergence is .6 per degree latitude" for ventrals, and on the same page they state that "The average number of ventrals gradually increases toward the north in *Tantilla gracilis* except in southern Oklahoma and central Arkansas (the area of intergradation between *T. g. gracilis* and *T. g. hallowelli*) where the change seems to be more abrupt." The data presented in Kirn *et al.* (1949) do not demonstrate an abrupt change.

The present sample of *T. gracilis* from Kansas is clearly within the geographic range of *T. g. hallowelli* as mapped by Kirn *et al.* (1949:241). However, for the sample from Kansas, the mean number of subcaudals of males is well within the diagnostic range for *T. g. gracilis* ([Table 6](#)), the mean number of subcaudals of females is closer to the mean of *gracilis* than it is to the mean of *hallowelli* ([Table 5](#)), and the mean of the ratio of tail length to total length of both sexes is within or very close to the diagnostic range for *gracilis* ([Table 6](#)).

TABLE 6. Comparison of the Means of Some Characteristics of the Sample of *Tantilla gracilis* from Kansas (Our Data) with Some of the Diagnostic Characteristics (Kirn *et al.*, 1949:240) of *Tantilla gracilis gracilis*.

CHARACTERISTIC	Diagnostic range for <i>T. g. gracilis</i>	Mean of the sample ("T. g. hallowelli") from Kansas
No. of subcaudals males	40-50	48.4 (N = 109)
Tail L./ Total L. females	.160-.200	.20 (N = 79)
Tail L./ Total L. males	.200-.235	.23 (N = 109)

Smith and Sanders (1952:218) pointed out an error in the range of subcaudals in female *T. g. gracilis* as stated in table 1 (p. 247) of Kirn *et al.* We add some additional corrections that should be made on the same page. In table two, column two ("Sex"), the words "male" and "female" are reversed for each characteristic. Also, in table two, column four ("Means from data of Force"), the means for caudals should be corrected from 33.61 to 43.61 in females and from 39.99 to 50.32 in males, as based on our recalculations from Force's data; we did not recalculate the means of Force's data for the other characteristics.

The discussion above demonstrates that the number of subcaudals and the ratio of tail length to total length are weak characteristics for diagnosing *T. g. gracilis* and *T. g. hallowelli*. The only remaining diagnostic characteristic given by Kirn *et al.* (1949) is the number of ventrals. Nevertheless, a step-cline in the number of ventrals, if one exists, is yet to be demonstrated. Therefore, we completely agree with Dowling (1957:32), who stated that "... the overlap in each character is so great that any subspecific differences appear to be as yet undefined. It is here suggested that *T. gracilis* be retained as binomial...."

LITERATURE CITED

BAIRD, S. F. and GIRARD, C.

1853. Catalogue of North American Reptiles in the museum of the Smithsonian Institution. Part I.—Serpents. Smithsonian Misc. Colls., 2:1-172.

BLANCHARD, F. N.

1938. Snakes of the genus *Tantilla* in the United States. Zool. Ser. Field Mus. Nat. Hist., 20(28):369-376.

CONANT, R.

1958. A field guide to reptiles and amphibians. Houghton Mifflin Co., Boston, 366 pp.

COPE, E. D.

1900. The crocodilians, lizards, and snakes of North America. U. S. Nat. Mus. Rept. (1898), 153-1270.

DOWLING, H. G.

1951. A proposed standard system of counting ventrals in snakes. British J. Herpetology, 1(5):97-99.

1957. A review of the amphibians and reptiles of Arkansas. Occas. Paps. Univ. Arkansas Mus., No. 3:1-51.

DUNN, E. R.

1928. A tentative key and arrangement of the American genera of Colubridae. Bull. Antivenin Inst. Amer., 2 (147):18-24.

FORCE, E. R.

1935. A local study of the opisthoglyph snake *Tantilla gracilis* Baird and Girard. Papers Michigan Acad. Sci., Arts and Letters, 20:645-659.

KIRN, A. J., BURGER, W. L., and SMITH, H. M.

1949. The subspecies of *Tantilla gracilis*. Amer. Midl. Nat., 42(1):238-251.

PETERS, J. A.

1960. The snakes of the subfamily Dipsadinae. Misc. Pubis. Mus. Zool., Univ. Michigan, No. 114:1-224.

1964. Dictionary of herpetology. Hafner Publ. Co., New York, 392 pp.

SCHMIDT, K. P. and DAVIS, D. D.

1941. Field book of snakes. G. P. Putnam's Sons, New York, 365 pp.

SMITH, H. M. and SANDERS, O.

1952. Distributional data on Texan amphibians and reptiles. Texas J. Sci., 4(2):204-219.

TAYLOR, E. H.

1936. Notes and comments on certain American and Mexican snakes of the genus *Tantilla*, with descriptions of new species. Trans. Kansas Acad. Sci., 39:335-348.

WRIGHT, A. H. and WRIGHT, A. A.

1957. Handbook of snakes. Comstock Publ. Assoc, Cornell, Vol. II, pp. 565-1105.

Department of Biology, The University of New Mexico, Albuquerque, 87106, and Department of Zoology, The University of Arizona, Tucson, 85721.

Transmitted July 11, 1967.

Textual representation of complete dorsal scale formula on [page 622](#).

$$\begin{array}{l} 15 \frac{6+7(2)}{6+7(2)} 13 \frac{+7(4)}{+7(5)} 14 \frac{\quad}{+7(5)} 15 \frac{6+7(11)}{6+7(11)} 13 \frac{\quad}{6=6+7(17)}, \\ 14 \frac{\quad}{6+7(18)} 13 \frac{\quad}{+7(25)}, \\ 14 \frac{+7(26)}{\quad} 15 \frac{\quad}{6+7(36)} 14 \frac{6+7(38)}{\quad} 13 \frac{+7(39)}{\quad} 14 \frac{\quad}{+7(42)}, \end{array}$$

$$15 \overline{-6(45)} 14 \overline{+7(53)},$$

$$15 \overline{7+8(67)} 14 \overline{+7(69)} 15 \overline{6+7(90)} 14 \overline{+7(93)} 15 \overline{6+7(99)},$$

$$14 \overline{+7(100)} 15(122).$$

*** END OF THE PROJECT GUTENBERG EBOOK
MORPHOLOGICAL VARIATION IN A POPULATION OF THE SNAKE,
TANTILLA GRACILIS BAIRD AND GIRARD ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

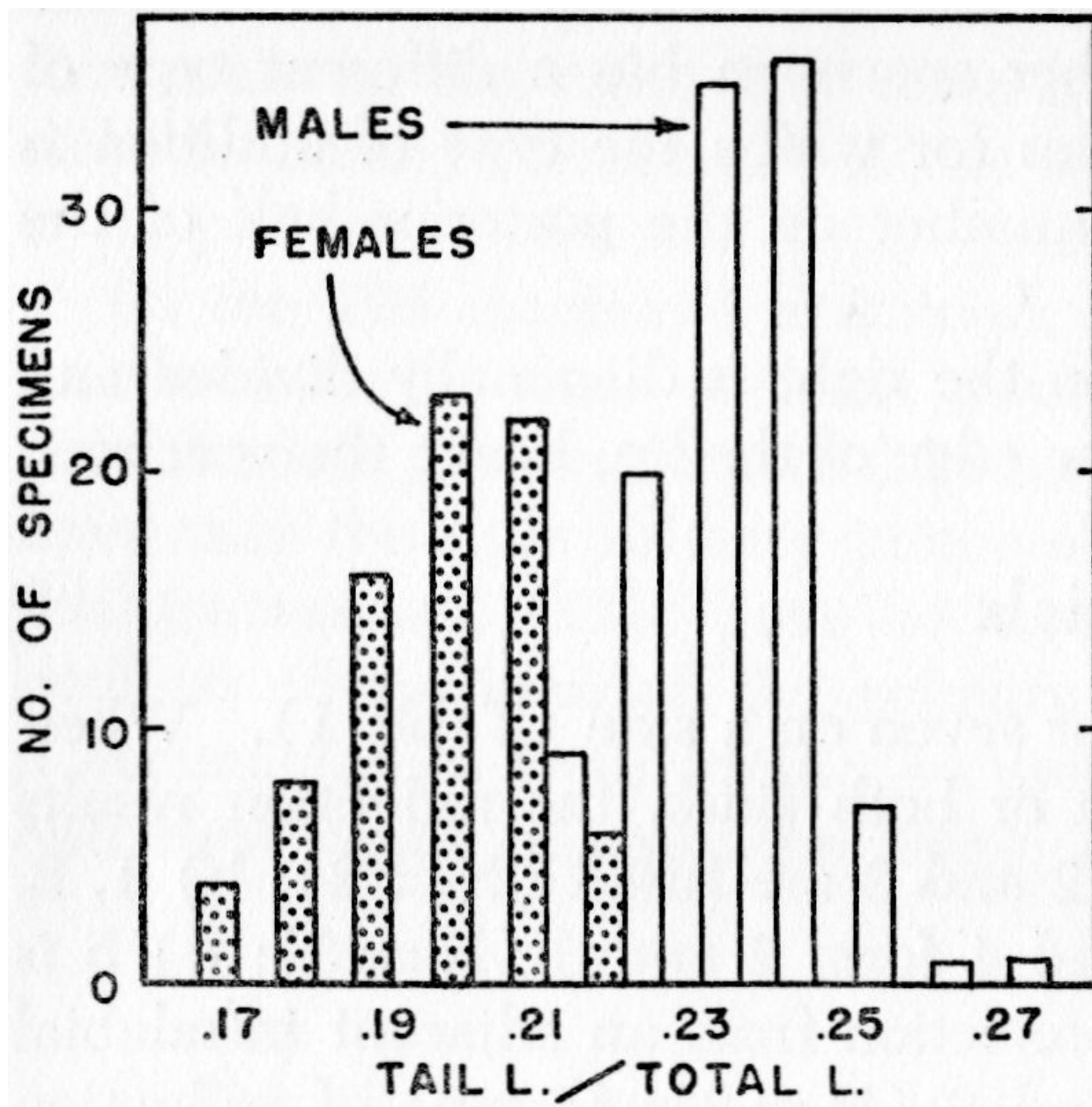
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

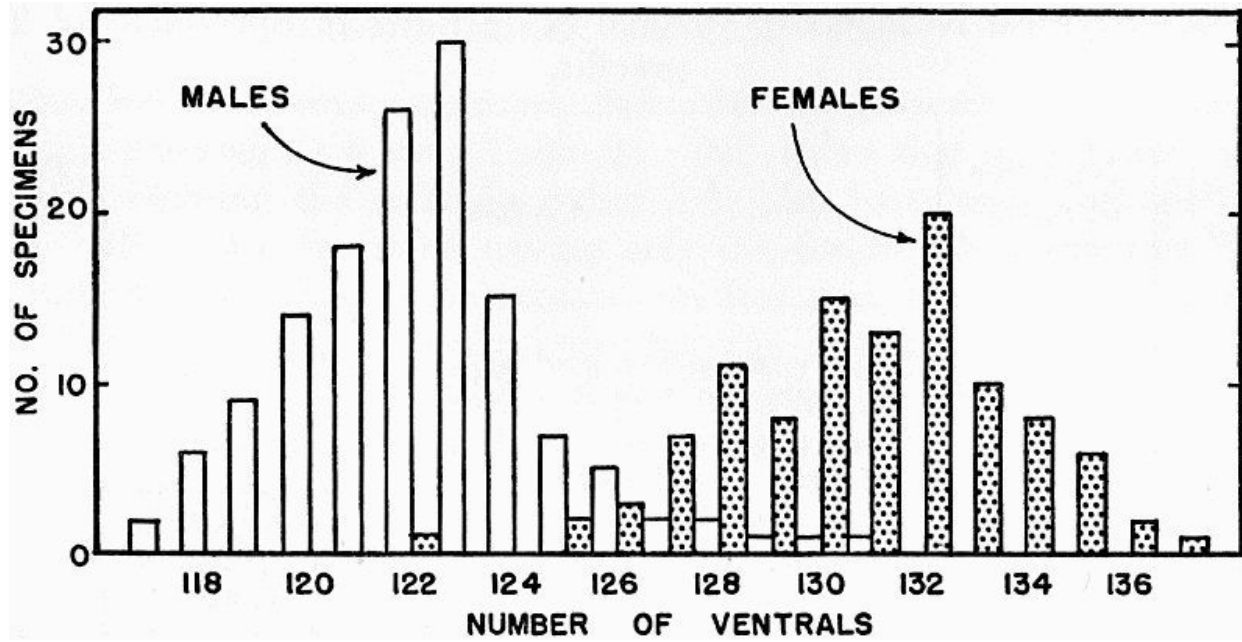
copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

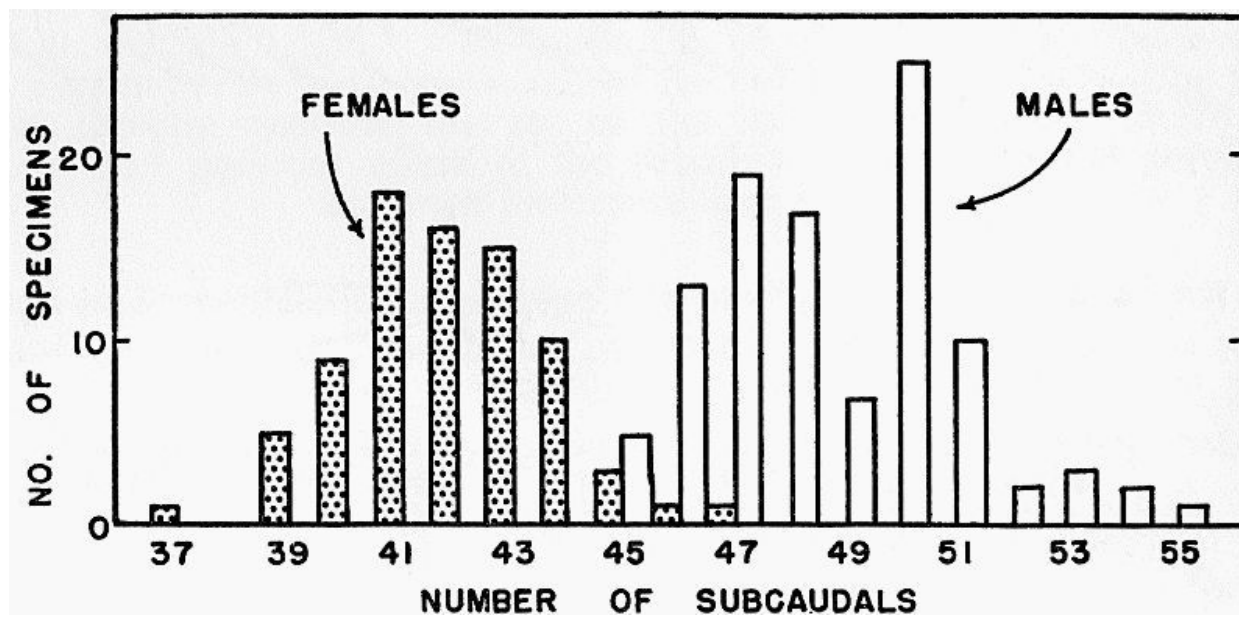
This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



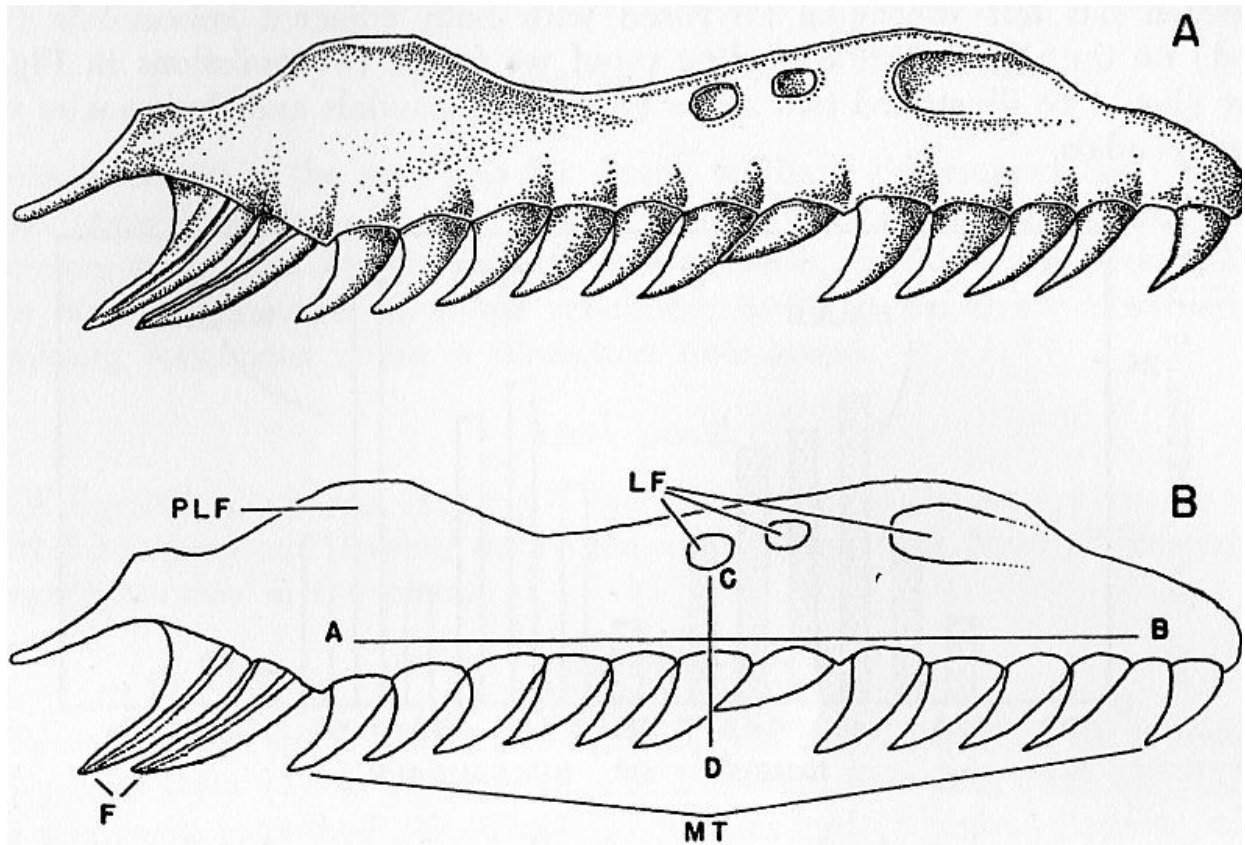
[back](#)



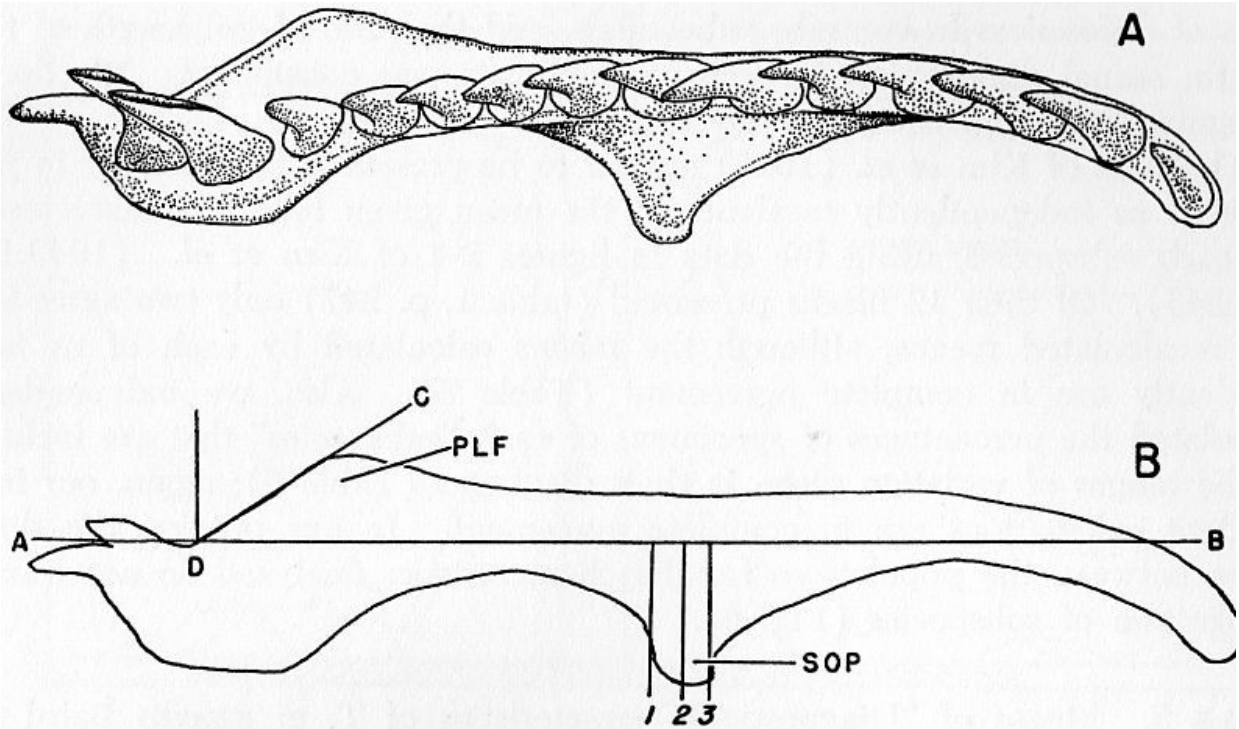
[back](#)



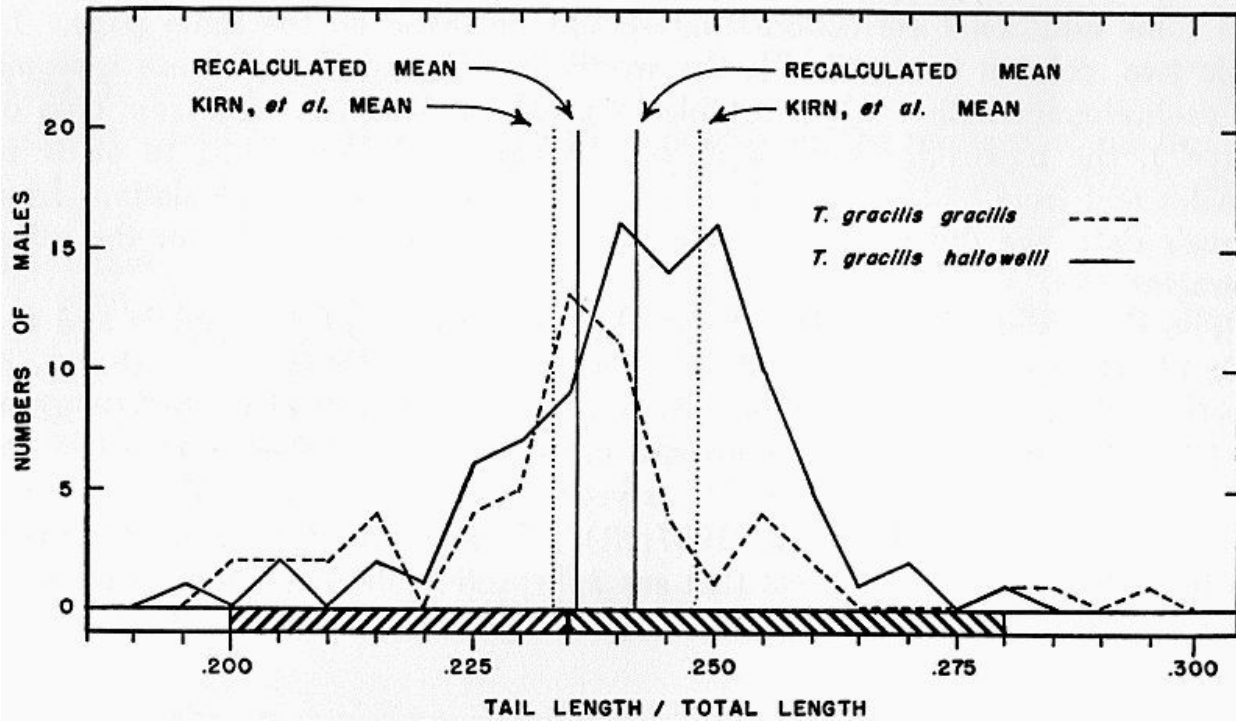
[back](#)



[back](#)



[back](#)



[back](#)